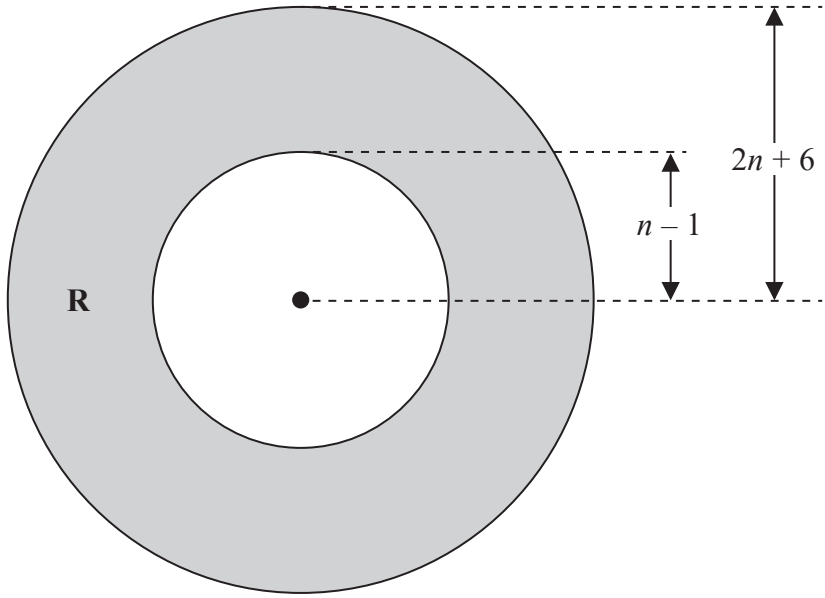
12 The region **R**, shown shaded in the diagram, is the region between two circles with the same centre.



The outer circle has radius $(2n + 6)$
The inner circle has radius $(n - 1)$
All measurements are in centimetres.

The area of **R** is greater than the area of a circle of radius $(n + 13)$ cm.

n is an integer.

Find the least possible value of n.
You must show all of your working.

$\pi(2n + 6)^2 - \pi(n - 1)^2 > \pi(n + 13)^2$.

1 mark
Circle areas

1 mark
Form inequality

×	2n	6
2n	$4n^2$	12n
6	12n	36

×	n	−1
n	n^2	−n
−1	−n	1

×	n	13
n	n^2	13n
13	13n	169

1 mark
Expand all brackets

Cancel π: $3n^2 + 26n + 35 > n^2 + 26n + 169$.

$2n^2 > 134$, so $n^2 > 67$.

1 mark
Isolate n^2

$n > 1; 8^2 = 64 < 67 < 81 = 9^2$, so least $n = 9$.

1 mark
Least integer9

(Total for Question 12 is 5 marks)